

Appendix: Detailed Derivations for the 2D Supersonic Marshak-Wave Model

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Scope

This appendix provides the detailed steps that are only summarized in the PRL manuscript: (i) the numerical closure of the 1D Hammer–Rosen model with a Marshak boundary, (ii) the radial eigenvalue problem in cylindrical geometry and its link to albedo, and (iii) the explicit wall-loss and wall-ablation accounting used in the 2D correction.

1 1D Core Equations and Marshak Closure

We use the standard power-law material model

$$\kappa^{-1} = gT^\alpha \rho^{-\lambda}, \quad e = fT^\beta \rho^{-\mu}, \quad (1)$$

and define

$$H(t) \equiv T_s^{4+\alpha}(t), \quad \varepsilon \equiv \frac{\beta}{4+\alpha}, \quad C \equiv \frac{16g\sigma_{\text{SB}}}{3f(4+\alpha)}\rho^{2-\mu-\lambda}. \quad (2)$$

The Hammer–Rosen front relation is

$$x_F^2(t) = \frac{2+\varepsilon}{1-\varepsilon} C H^{-\varepsilon}(t) \int_0^t H(t') dt'. \quad (3)$$

The material energy per unit cross-sectional area is

$$E(t) = f\rho^{1-\mu}(1-\varepsilon)x_F(t)H^\varepsilon(t), \quad (4)$$

with boundary flux condition

$$\dot{E}(t) = F(0, t). \quad (5)$$

At the entrance boundary, the Marshak condition is

$$\sigma_{\text{SB}}T_D^4(t) = \sigma_{\text{SB}}T_s^4(t) + \frac{1}{2}F(0, t). \quad (6)$$

Equations (3)–(6) close the 1D problem.

1.1 Implicit time-marching form for $T_s(t)$

Following the Appendix-A construction used in the long manuscript, combine Eqs. (3) and (4) to get

$$E^2(t) = Z_1 H^\varepsilon(t) \int_0^t H(t') dt', \quad Z_1 \equiv f^2 \rho^{2(1-\mu)} (2 + \varepsilon)(1 - \varepsilon) C. \quad (7)$$

At time step $t_n = t_{n-1} + \Delta t$, define $H_n \equiv H(t_n)$ and

$$I_{n-1} \equiv \int_0^{t_{n-1}} H(t') dt'. \quad (8)$$

Using a trapezoid approximation for the last interval,

$$\int_{t_{n-1}}^{t_n} H(t') dt' \approx \frac{H_{n-1} + H_n}{2} \Delta t, \quad (9)$$

Eq. (7) becomes one nonlinear equation for H_n :

$$Z_1 \left(I_{n-1} + \frac{H_{n-1} + H_n}{2} \Delta t \right) H_n^\varepsilon - E_n^2 = 0. \quad (10)$$

The input energy is advanced from the Marshak flux:

$$F_n = 2\sigma_{\text{SB}} (T_{D,n}^4 - T_{s,n-1}^4), \quad E_n = E_{n-1} + \frac{F_n + F_{n-1}}{2} \Delta t. \quad (11)$$

Then solve Eq. (10) for the positive root $H_n > 0$, recover

$$T_{s,n} = H_n^{1/(4+\alpha)}, \quad (12)$$

and update

$$I_n = I_{n-1} + \frac{H_{n-1} + H_n}{2} \Delta t, \quad x_{F,n}^2 = \frac{2 + \varepsilon}{1 - \varepsilon} C H_n^{-\varepsilon} I_n. \quad (13)$$

This is the practical algorithm used to compute the 1D centerline front with Marshak boundary forcing.

2 Wall-Energy Loss: Full Integral Form

Let $t_0(x)$ be the arrival time of the front at axial location x , defined by

$$x_F(t_0(x)) = x. \quad (14)$$

After arrival, the local wall element is exposed for $\Delta t_x = t - t_0(x)$. For wall material m (Au/Cu/Be), the absorbed areal energy is written in self-similar form

$$E_{w,m}(\Delta t_x; T_s) \approx A_m T_s^{p_m}(t_0) \Delta t_x^{q_m}, \quad \Delta t_x > 0. \quad (15)$$

For the Au case used in the paper, one common fit is

$$E_{w,\text{Au}}(\Delta t_x) \simeq 0.59 T_0^{3.35} \Delta t_x^{0.59} \quad [\text{hJ/mm}^2], \quad (16)$$

with coefficients from the cited subsonic wall-heating model.

The total wall-loss energy is then

$$E_W(t) = 2\pi \int_0^{x_F(t)} R(x, t) E_{w,m}(t - t_0(x)) dx. \quad (17)$$

In differential form, this is equivalent to

$$\dot{E}_W(t) = 2\pi \int_0^{x_F(t)} R(x, t) \dot{E}_{w,m}(t - t_0(x)) dx. \quad (18)$$

Numerically, Eqs. (17)–(18) are evaluated on the same front-time grid used for $x_F(t)$, with interpolation for $t_0(x)$.

3 Wall Ablation and Density Feedback

For subsonic wall response (relevant for Au and Cu sleeves in our regime), the inward wall velocity is modeled as

$$u_w(\Delta t_x; T_s) \approx B_m \tilde{u}(\rho_w) T_s^{\eta_m}(t_0) \Delta t_x^{\xi_m}. \quad (19)$$

For Au, a representative fit is

$$u_{w,\text{Au}} \simeq 510.1 \tilde{u}(\rho_{\text{Au}}) T_0^{0.716} \Delta t_x^{0.036} \quad [\text{km/s}]. \quad (20)$$

The effective tube radius follows

$$R(x, t) = R_0 - \int_{t_0(x)}^t u_w(t' - t_0(x)) dt'. \quad (21)$$

The foam volume inside the heated region is

$$V(t) = \int_0^{x_F(t)} \pi R^2(x, t) dx, \quad (22)$$

and the mean density update is

$$\rho(t) = \rho_0 \frac{V_0}{V(t)}. \quad (23)$$

Because $C \propto \rho^{2-\mu-\lambda}$, this introduces a density-history correction in the effective front coefficient (as used in the implementation).

4 Radial Eigenvalue Problem and Albedo Coupling

5 Simulation details

The simulation introduced in the letter is a 2D diffusion simulation without hydrodynamics, using cylindrical geometry. The equations discretization is done through finite differencing on the following coupled equations:

$$\frac{\partial E}{\partial t} = \frac{1}{r} \frac{\partial}{\partial r} \left(r D \frac{\partial E}{\partial r} \right) + \frac{\partial}{\partial z} \left(D \frac{\partial E}{\partial z} \right) - \chi c \sigma (E - U). \quad (24)$$

This is the form we use throughout the derivation below.

5.1 Material Energy Equation

The material energy density $U(r, z, t)$ evolves according to:

$$\frac{1}{\beta} \frac{\partial U}{\partial t} = \chi c \sigma (E - U), \quad (25)$$

where $\beta = \frac{\partial U_R}{\partial U_m}$ is the ratio of radiation to material heat capacity (dimensionless coupling strength).

5.2 Decomposition of the Diffusion Operator

Define the space-dependent diffusion coefficient:

$$D(r, z, t) \equiv \frac{c}{3\sigma(T(r, z, t))}. \quad (26)$$